

HIGH-INTENSITY INTERVAL TRAINING VS. REPEATED-SPRINT TRAINING IN TENNIS

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ABSTRACT

Fernandez-Fernandez, J, Zimek, R, Wiewelhoe, T, and Ferrauti, A. High-intensity interval training vs. repeated-sprint training in tennis. *J Strength Cond Res* 26(1): 53–62, 2012—The aim of this study was to compare the effects of high-intensity interval training (HIIT) and repeated-sprint training (RST) on aerobic fitness, tennis-specific endurance, linear and repeated-sprint ability (RSA), and jumping ability. Thirty-one competitive male tennis players took part in a training intervention of 6 weeks. The players were matched into 3 groups, HIIT ($n = 11$), RST ($n = 12$), or control group (CON, $n = 9$). The results showed significant time $\times$ intervention interactions for $\dot{V}O_{2\text{peak}}$, with a significant increase in the $\dot{V}O_{2\text{peak}}$ level of 6.0% in HIIT ($p = 0.008$) and 4.9% in RST ($p = 0.010$), whereas no changes occurred in CON. However, the following differences were found between the intervention groups: The HIIT-induced greater improvements in tennis-specific endurance (HIIT 28.9% vs. RST 14.5%; $p < 0.05$) and RST led to a significant improvement in RSA (i.e., reduction in the mean sprint time of 3.8%; $p < 0.05$). Neither training strategy induced any effects on jumping and sprinting abilities. Both training interventions showed similar improvements in general aerobic fitness. Also, the present results suggest that RST represents a time-efficient stimulus for a simultaneous improvement of general and tennis-specific aerobic fitness as well for RSA.

KEY WORDS high intensity training, repeated-sprint ability, specific endurance, aerobic capacity

INTRODUCTION

Tennis involves intermittent, high-intensity efforts interspersed with periods of low-intensity activity, during which active recovery (between points) and passive periods (between changeover breaks in play) take place (20), over an extended period of time (i.e., in some cases >5 hours) (15,20,34,47). In this situation, competitive

tennis players need a mixture of fitness qualities such as speed, agility, and power combined with a well-developed aerobic fitness to achieve high levels of performance (34).

During match play, demands alternate between energy provision for bouts of high-intensity work (e.g., several strokes, quick changes of direction, short accelerations, and decelerations), via intramuscular phosphates and glycolysis, and replenishing energy sources and restoring homeostasis during the intervals in between (by oxidative metabolism) (20,25,47,48). Thus, it seems that the training of competitive players should focus on improving their ability to repeatedly perform high-intensity exercise and to recover rapidly from it (20). For these reasons, tennis training should include physical exercise aimed to enhance both aerobic and anaerobic fitness.

Several studies from team sports, mostly football, have shown that high-intensity interval training (HIIT) (i.e., work and rest intervals ranging from 15 seconds to 4 minutes; 90–100% velocity at the level of maximum oxygen uptake $\dot{V}O_{2\text{max}}$; heart rate [HR] values $>90\%$ of maximum HR [HRmax]; work-to-rest ratios of 1:1–4:1) (37) is an effective training strategy to enhance the aerobic capacity without negatively affecting strength, power, or sprint performance (29,31). This kind of training elicits increases in cardiovascular parameters such as heart size, blood flow capacity, and artery distensibility (35,43), thus improving the capacity of the cardiovascular system to transport oxygen, resulting in faster muscle and pulmonary $\dot{V}O_2$ kinetics and higher $\dot{V}O_{2\text{max}}$ (35,38,39). Hence, a greater amount of energy can be supplied aerobically, allowing a player to both sustain intense exercise for longer durations and also recover more rapidly between high-intensity phases of the game (28,30).

Speed endurance and repeated-sprint ability (RSA)-based training is characterized by performing repeated sprints with minimal recovery between sprint bouts (i.e., 10–20 maximal sprints or shuttle sprints of ≤ 10 seconds, with brief recovery periods (≤ 60 seconds); work:rest ratio of 1:4–1:6) (25,48). During such training, there is an increase in the activity of some anaerobic enzymes, which leads to a higher rate of anaerobic energy turnover and increases the number of muscle membrane transport proteins involved in pH regulation and muscle capillarization and in some cases enhances the muscle buffering capacity (18,19,26). Also, the performance of maximal or near-maximal short-term efforts

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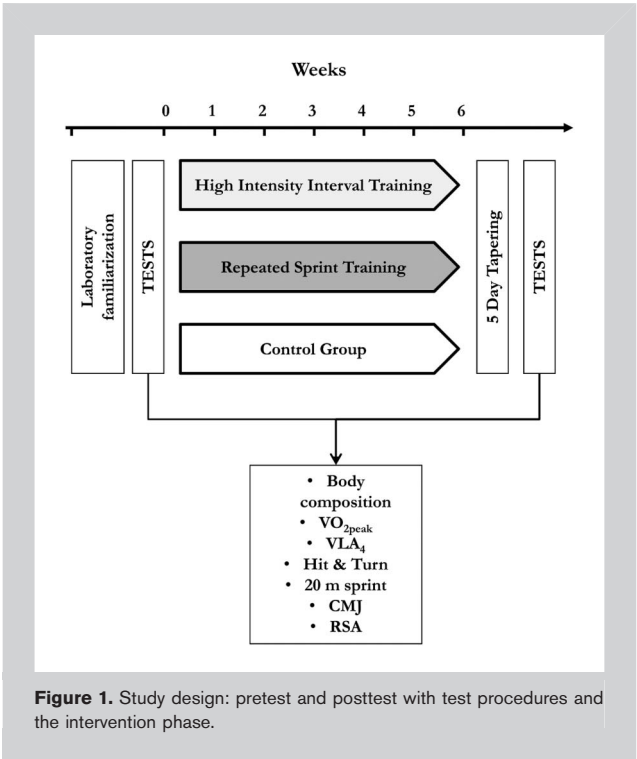
can lead to higher $\dot{V}O_{2\max}$ values and an increased aerobic enzyme activity (11–13,22,23). These findings suggest the effectiveness of both training systems (i.e., repeated-sprint and interval training) for enhancing both the aerobic and anaerobic capacities and, therefore, could be considered as appropriate training strategies for tennis players.

Despite the growing effectiveness from the above-mentioned training strategies for team sports (30), to the authors' knowledge, no studies have focused on the specific effects of different high-intensity training programs in tennis players. Therefore, the aim of this study was to compare the effects of repeated-sprint training (RST) with an HIIT on aerobic fitness, tennis-specific endurance, linear sprint ability and RSA, and jumping ability in tournament tennis players.

METHODS

Experimental Approach to the Problem

A randomized, controlled, and longitudinal (i.e., pretest-posttest) design was used. Before any baseline testing, all the participants attended a laboratory familiarization visit to introduce the testing or training procedures and also to ensure that any learning effect was minimal for the baseline measures. The players completed baseline tests for body composition, peak $\dot{V}O_2$ ($\dot{V}O_{2peak}$), velocity achieved at a blood lactate concentration of 4 mmol·L⁻¹ (vLA₄), specific endurance field test (i.e., Hit and Turn test), 20-m sprint, countermovement jump (CMJ), and RSA. The players were matched into 3 groups according to their laboratory measurements and completed HIIT (*n* = 11), RST (*n* = 12), or served as control group (CON, *n* = 9). The training intervention consisted of 1 week of tests (pretest), 6 weeks of supervised training intervention (3 times per week) in addition to 1–2 self-regulated moderate- to low-intensity injury prevention (e.g., core training, shoulder strengthening and flexibility) sessions during the week, 5 days of tapering and 1 week of tests (posttest) (Figure 1). No additional strength, power, or plyometric training was completed. The control group followed their normal tennis training (3 × week), in addition to 1–2 self-regulated moderate- to low-intensity injury prevention (e.g., core training, shoulder strengthening, and flexibility) sessions. To reduce the interference of uncontrolled variables, all the subjects were instructed to maintain their habitual lifestyle and normal dietary intake before and during the study. All the tests and training sessions were performed on an indoor tennis court (i.e., greenset surface) where ambient temperature ranged from 18 to 22°C. The subjects were told not to exercise on the day before a test and



to consume their last (caffeine-free) meal at least 3 hours before the scheduled test time.

Subjects

A total of 31 competitive male tennis players (International Tennis Number 3 [ITN]) involved in regular tennis competition at the national level, with a national ranking between 150 and 250, volunteered to take part in this study (Table 1). The mean training background of the players was 12.0 ± 3.6 years, which focused on tennis-specific training (i.e., technical and tactical skills), aerobic and anaerobic training (i.e., on- and off-court exercises), and strength training. Before any participation, the experimental

TABLE 1. Subjects' characteristics.*†

	HIIT (<i>n</i> = 11)	RST (<i>n</i> = 12)	CON (<i>n</i> = 9)
Age (y)	22.6 ± 4.8	21.2 ± 5.1	22.1 ± 3.3
Weight (kg)	79.6 ± 10.8	73.5% ± 8.1	76.5 ± 6.6
Height (cm)	182.9 ± 7.3	178.0 ± 5.9	180.5 ± 4.8
Fat percentage (%)	14.1 ± 2.7	14.2 ± 2.9	14.6 ± 1.9
$\dot{V}O_{2peak}$ (ml·kg ⁻¹ ·min ⁻¹)	56.3 ± 4.0	55.6 ± 5.0	57.7 ± 3.9

*HIIT = high intensity interval training group; RST = repeated-sprint training group; CON = control group.
†Data are mean ± SD.

procedures and potential risks were explained fully to the subjects, and they all provided written informed consent. The study was performed in accordance with the ethical standards reported by Harriss and Atkinson (27) and conformed to the recommendations of the Declaration of Helsinki.

Procedures

Laboratory Measurements. An incremental treadmill test (Ergo ELG 2, Woodway GmbH; Germany) was used to determine $\dot{V}O_{2\text{peak}}$ using an online breath-by-breath gas collection system (ZAN600USB, Germany). The test began with an initial velocity of $8 \text{ km}\cdot\text{h}^{-1}$, increasing $2 \text{ km}\cdot\text{h}^{-1}$ every 3 minutes until subjective exhaustion, with a constant grade of 1.5% (49). Expired air was continuously analyzed for gas volume, O_2 concentration, and CO_2 concentration. The volume calibration of the system was conducted before each test day, and the gas calibration was performed before each test using instructions provided by the manufacturer. The mean of the 5 highest $\dot{V}O_2$ values obtained during the test was defined as the $\dot{V}O_{2\text{peak}}$. As other performance criteria, we reported the maximum velocity achieved at a blood lactate concentration of $4 \text{ mmol}\cdot\text{L}^{-1}$ (vLA_4) (41). The HR was monitored and recorded at 5-second intervals during the exercise (S610, Polar Electro, Kempele, Finland), and HRmax was determined as the highest 5-second mean value. Capillary blood samples were taken from hyperemized earlobe (Finalgon®) during the 30-second break immediately after finishing each velocity level and at the time point of exhaustion. Blood samples were hemolyzed in 2-ml microtest tubes and analyzed enzymatic amperometrically by the Biosen S-Line Sport (EKF-Diagnostik, Germany) immediately after each test (22). The rating of perceived exertion (RPE) was obtained using the 6- to 20-category RPE scale (5). The scale was explained before the exercise. The subjects were asked: "How hard do you feel the exercise was?" after finishing each velocity level.

Hit and Turn Tennis Test. The Hit and Turn Test test was developed as an acoustically controlled progressive on-court fitness test for tennis players, which can be performed simultaneously by one or more players (Figure 2) (22). The test involves specific movements along the baseline (i.e., side steps and running), combined with forehand and backhand stroke simulations at the doubles court corner (distance 11.0 m). At the beginning of each test level, the players stand with their racket in a frontal position in the middle of the baseline. Upon hearing a signal, the player turns sideways and runs to the prescribed (i.e., by the CD player) backhand or forehand corner. After making their shot, they return to the middle of the court using side steps or crossover steps (while looking at the net). When passing the middle of the baseline again, they turn sideways and continue to run to the opponent's opposite corner. The end of the test was considered when players fail to reach the cones in time or was no longer able to fulfill the

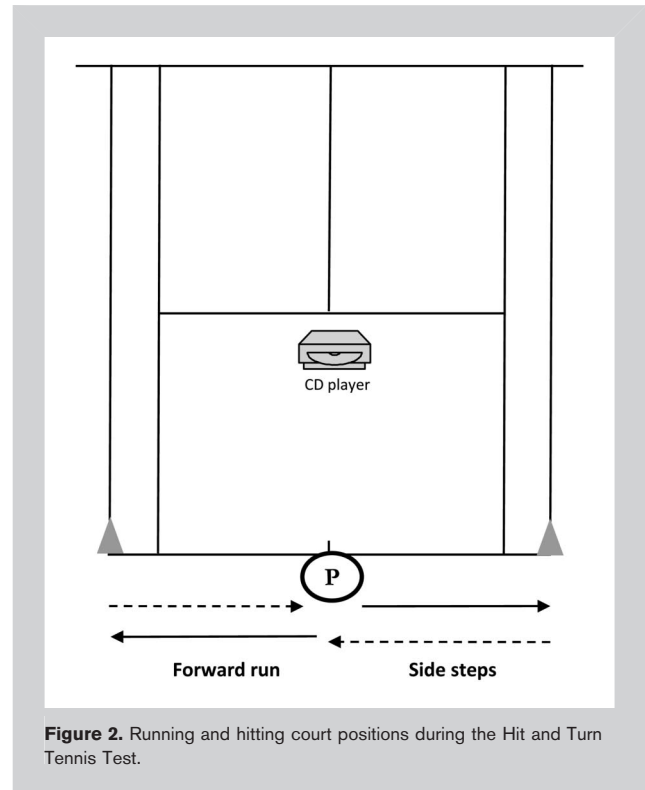


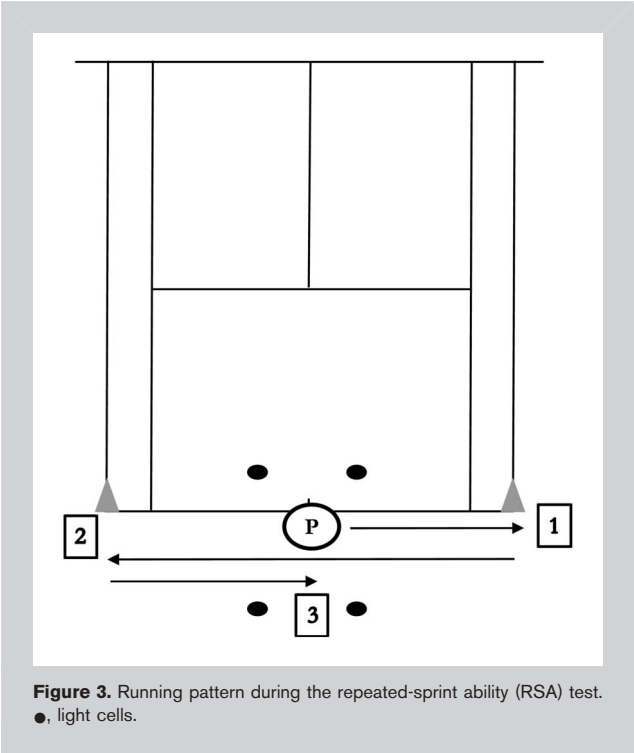
Figure 2. Running and hitting court positions during the Hit and Turn Tennis Test.

specific movement pattern. Maximal completed level was used for the determination of the tennis-specific endurance capacity (22). The HR (S610, Polar Electro) was measured continuously until exhaustion, and the RPE was obtained using the 6- to 20-category RPE scale at the beginning and at the end of the test.

Vertical Jumping. The CMJs without arm swing was performed on a contact platform (Haynl Elektronik, Germany) according to Bosco et al. (6). Each player performed 3 maximal CMJs interspersed with 45 seconds of passive recovery, and the best height for each was recorded.

Twenty-meter Sprint Run. The velocity during a 20-m dash in a straight line was measured by means of photocell gates placed 1.0 m above the ground level (Sportronic TS01-R04, Leutenbach-Nellmersbach, Germany). Each sprint was initiated from an individually chosen standing position, 50 cm behind the photocell gate, which started a digital timer. Each player performed 3 maximal 20-m sprints interspersed with 3 minutes of passive recovery, and the fastest time achieved was recorded.

Repeated-Sprint Ability Shuttle Test. To measure RSA, we used a test consisting of ten 22-m shuttle sprints (i.e., 5 m + 11 m + 5 m) (Figure 3). This test was designed to measure both repeated sprint and change in direction abilities. The player stood with his racket in a frontal position (i.e., looking to the net) in the middle of the baseline. Upon hearing a signal, players turned sideward and ran to the prescribed backhand (left) or



forehand (right) corner. The players were instructed to run forward in a straight line and turn 180° when their feet were in line with the turning point marked with a cone and to touch it with the racket. After touching the first cone with the racket, they returned to the opposite side of the court running forwards, touching the second cone and turning to run to the initial position. After 15 seconds of passive recovery, the players restarted again. Five seconds before the start of each sprint, the subjects assumed the ready position and waited for the start signal. Each shuttle sprint time was measured using a photocells system (Sportronic TS01-R04, Leutenbach-Nellmersbach, Germany), and the mean time and percent decrement during the RSA test were calculated. Each player completed a preliminary single shuttle sprint test, which was

used as the criterion score during the subsequent shuttle sprint test. After the first preliminary single shuttle sprint, the subjects rested for 5 minutes before the start of the RSA test. If performance in the first sprint of the RSA test was worse than the criterion score (i.e., an increase in time >2.5%), the test was immediately terminated, and the subjects were required to repeat the RSA test with maximum effort after a 5-minute rest (21,48).

Training Program. The training program took place during the preseason period. Throughout the study, the subjects trained on 3 d·wk⁻¹ over a period of 6 weeks, resulted in 18 training sessions. Both training interventions were performed on-court, separated by at least 48 hours. During each session, 3 intensive exercise bouts (i.e., HIIT or RST) were interrupted by an on-court tennis game (e.g., “2 against one player” [2:1 game]), where 2 players took turns in playing against the single one after each point. During each training session, each player played one time alone. The intensity of the 2:1 game was fixed between 75 and 85% of HR_{max} (Table 2).

For the HIIT, the program consisted of replicating the movements executed during the Hit and Turn Test (i.e., side steps and running, combined with forehand and backhand stroke simulations; see Figure 2). The speed of running was controlled via beep signals from a CD, the stroke production being exactly simultaneous to the beep signals coming from it. The players performed 3 sets of 3- to 90-second runs at an intensity of the maximum level reached in the Hit and Turn Test. The intensity was individually adapted during the 6-week period considering the HR being above 90–95% of HR_{max} during the exercise (i.e., 710-second work, 540-second rest). Each set was separated by 180 seconds of active recovery, running around the court at an intensity of 70% of HR_{max}, and 8 minutes of an on-court tennis game (i.e., 2:1 game). For the RST, the program consisted of replicating the RSA shuttle test (Figure 3), performing 3 sets of 10 22-m shuttle sprints, separated by 20 seconds of passive recovery between repetitions (i.e., 50-second work, 150-second rest), and 8 minutes of an on-court tennis game (i.e., 2:1 game) between sets.

TABLE 2. Schematic representation of the training intervention.

Repeated-sprint training		High-intensity interval training
10 × ±5 s all-out, 15-s rest Game 2:1	First set	3 × 90-s level _{max} *, 3-min active recovery (70% of HR _{max}) Game 2:1
10 × ±5 s all-out, 15-s rest Game 2:1	Second set	3 × 90-s level _{max} *, 3-min active recovery (70% of HR _{max}) Game 2:1
10 × ±5 s all-out, 15-s rest Game 2:1	Third set	3 × 90-s level _{max} *, 3-min active recovery (70% of HR _{max}) Game 2:1

*Maximum level reached in the Hit and Turn Tennis Test.

During both training sessions, the players were equipped with a short-range telemetry system, which allowed the measurement of HR (Polar S610). The intensity was controlled on-line telemetrically (Polar Windlink), and the data obtained from the HR monitors were downloaded on a portable PC and classified based on percentage time spent in 5 zones: (a) <60% HRmax; (b) 61–70% HRmax; (c) 71–80% HRmax; (d) 81–90% HRmax; and (e) >91% HRmax (1). Individual average duration and intensity of these periods were calculated. Capillary blood samples were taken from hyperemized earlobe (Finalgon®) immediately after finishing each set, and, randomized, during the whole training period. Blood samples were hemolyzed in 2-ml microtest tubes and analyzed enzymatic amperometrically by the Biosen S-Line Sport (EKF-Diagnostik, Germany) immediately after each training session.

Statistical Analyses

Data are reported as mean $\pm$ SD. After the pretests, the $\dot{V}O_{2\text{peak}}$ of the experimental groups was compared using an independent sample *t*-test to ensure that they started the experiment with a similar aerobic fitness capacity. Before using parametric tests, the assumption of normality was verified. A 2-way analysis of variance (ANOVA) for repeated measurements was used to determine the differences among tests and between groups. The independent variables included one between-subject factor (3 levels of training intervention, HIIT, RST, and CON), and one within-subject factor time with 2 levels (pretest and posttest). We used the ANOVA to test the null hypothesis of no difference in changeover time in response to the training intervention (main effect for time) and the null hypothesis of no difference in the changeover time between training groups (training intervention $\times$ time interaction). To allow a better

interpretation of the results, effect sizes were also calculated (partial eta squared, η^2). Values of 0.01, 0.06, and >0.15 were considered small, medium, and large, respectively. Training load, HR, blood lactate, and perception of muscle pain between the 2 intervention groups was analyzed using an independent samples *t*-test. The SPSS statistical software package (version 15, SPSS Inc., Chicago, IL, USA) was used for statistical calculations. The level of statistical significance was set at $p \leq 0.05$.

RESULTS

Laboratory Measurements

The physiological variables analyzed ($\dot{V}O_{2\text{peak}}$, vLA_4 , HRmax) at baseline and after 6 weeks of intervention are shown in Table 3. There was a significant group $\times$ time interaction in $\dot{V}O_{2\text{peak}}$ levels ($p = 0.040$) and also significant main time effects for $\dot{V}O_{2\text{peak}}$ and vLA_4 , with a significant increase in the $\dot{V}O_{2\text{peak}}$ level of 4.9% ($p = 0.010$) and 6.0% ($p = 0.008$), for both HIIT and RST, respectively. There were no significant differences in the $\dot{V}O_{2\text{peak}}$ levels for the CON (−0.4%; $p = 0.879$), being significantly lower than values reported for HIIT and RST. The HRmax did not change from pretesting to posttesting ($p = 0.091$). Moreover, there was no significant group $\times$ time interaction, for neither HRmax nor vLA_4 in all the groups analyzed (RST, HIIT, or CON) ($p > 0.05$). Regarding the average caloric expenditure for both training protocols, the players involved in the HIIT burnt $922.4 \pm 80.4 \text{ kcal} \cdot \text{min}^{-1}$, which was significantly higher than caloric expenditure for RST ($922.4 \pm 80.4 \text{ kcal} \cdot \text{min}^{-1}$) ($p = 0.006$).

Hit and Turn Tennis Test

A significant group $\times$ time interaction ($p = 0.001$) was found in the Hit and Turn Test performance (Table 4). Also, the

TABLE 3. Effects of high-intensity interval training and repeated-sprint training on laboratory measurements.*†

Variable	Group	Pretest	Posttest	Time		Interaction	
				p	η^2_{partial}	p	η^2_{partial}
$\dot{V}O_{2\text{peak}}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	HIIT	56.3 ± 4.0	$59.1 \pm 2.9^{\ddagger}\S$	0.001	0.328	0.040	0.17
	RST	55.6 ± 5.0	$58.6 \pm 2.9^{\ddagger}\S$				
	CON	57.3 ± 4.0	57.4 ± 3.8				
vLA_4 ($\text{km} \cdot \text{h}^{-1}$)	HIIT	13.1 ± 0.6	13.8 ± 1.0	0.008	0.228	0.875	0.009
	RST	13.4 ± 2.1	14.2 ± 0.9				
	CON	12.9 ± 1.1	13.1 ± 1.3				
HRmax ($\text{b} \cdot \text{min}^{-1}$)	HIIT	188.9 ± 6.6	187.3 ± 8.6	0.091	0.099	0.885	0.009
	RST	196.6 ± 10.4	194.8 ± 11.7				
	CON	194.9 ± 8.7	191.8 ± 5.6				

* $\dot{V}O_{2\text{peak}}$ = peak oxygen uptake; vLA_4 = velocity at a lactate concentration of 4 $\text{mmol} \cdot \text{L}^{-1}$; HRmax = maximum heart rate; HIIT = high-intensity interval training; RST = repeated-sprint training; CON = control group.

†Data are mean $\pm$ SD.

‡Significant differences between pre and posttraining.

§Significantly different from CON.

TABLE 4. Effects of high-intensity interval training and repeated-sprint training on performance variables analyzed.*†

Variable	Group	Pre	Post	Time		Interaction	
				<i>p</i>	η^2_{partial}	<i>p</i>	η^2_{partial}
RSA (s)	HIIT	5.2 ± 0.1	5.2 ± 0.1‡	0.002	0.293	0.001	0.402
	RST	5.3 ± 0.2	5.1 ± 0.2‡§				
	CON	5.3 ± 0.3	5.3 ± 0.3				
RSA decrement (s)	HIIT	0.4 ± 0.1	3.1 ± 0.1	0.676	0.006	0.230	0.100
	RST	0.4 ± 0.1	3.2 ± 0.1				
	CON	0.4 ± 0.2	3.2 ± 0.1				
HTT (level _{max})	HIIT	14.0 ± 3.2	18.0 ± 1.4‡§	0.000	0.548	0.001	0.363
	RST	14.0 ± 3.0	16.1 ± 1.8‡§				
	CON	15.2 ± 2.3	15.5 ± 2.2				
CMJ (cm)	HIIT	41.0 ± 3.0	40.4 ± 3.0	0.265	0.088	0.015	0.188
	RST	38.4 ± 4.0	38.0 ± 3.3				
	CON	42.5 ± 4.7	40.5 ± 4.5§				
20-m Sprint (s)	HIIT	3.1 ± 0.1	3.1 ± 0.1	0.411	0.059	0.266	0.042
	RST	3.2 ± 0.1	3.2 ± 0.1				
	CON	3.2 ± 0.1	3.2 ± 0.1				

*RSA = repeated-sprint ability test; HTT = Hit and Turn Tennis Test; HIIT = high-intensity interval training; RST = repeated-sprint training; CON = control group; CMJ = countermovement jump.

†Data are mean ± SD.

‡Significantly different from CON.

§Significant differences between pretraining and posttraining.

||Significantly different from RST.

main effect for time was significant ($p = 0.000$). There was a significant increase in the maximum level achieved in the Hit and Turn Test for both groups, RST and HIIT (+14.5% [$p = 0.014$] and +28.9% [$p = 0.000$], respectively) from pretest to posttest, with significantly higher posttraining values for HIIT compared with RST ($p = 0.010$). The players from the CON had a fractional but no significant increase of 2.4% ($p = 0.549$).

Sprint and Jump Test

There was a significant main effect for time in CMJ performance ($p = 0.015$). No group $\times$ time interaction was found for jumping ($p = 0.265$) and sprinting performance ($p = 0.411$). Similarly, no pretest to posttest changes were found in 20-m sprint time ($p = 0.266$).

Repeated-Sprint Ability Test

Significant group $\times$ time interactions ($p = 0.001$) were found in the RSA mean time but not in RSA decrement (pretest: 0.36 ± 0.1 ; posttest: 0.31 ± 0.1 ; $p = 0.361$). The mean time

during the RSA test was significantly reduced in the RST (3.8%; $p = 0.000$) after the intervention, whereas there were no differences between pretest and posttest for the HIIT ($p = 0.951$) and CON ($p = 0.541$).

The mean ($\pm$ SD) HR of the players obtained in this study was 149 ± 6 and 152 ± 10.8 b $\cdot$ min $^{-1}$ for the HIIT and RST,

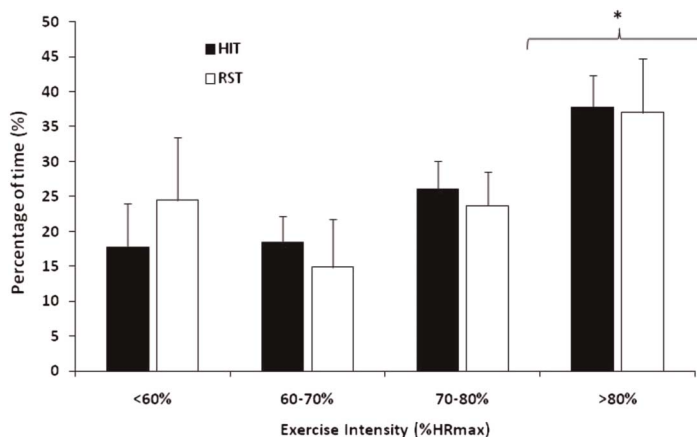


Figure 4. Percentage of time spent by players in the different heart rate (HR) categories during high-intensity interval and RSA-based training sessions. HIIT = interval training group; RSA = repeated-sprint ability; RST = repeated-sprint training group. *Significantly higher than any other average intensity for both training sessions.

respectively, with no significant differences between groups ($p = 0.47$). Figure 4 shows the percentage of time spent by the players in the different HR categories during the training sessions performed during the study. There were no significant differences between the HIIT and RST for any of the HR categories. The players spent 37.8 and 37% of the total time at exercise intensities $>80\%$ of HRmax, during HIIT and RST, respectively, which was significantly higher ($p < 0.05$) than any other average intensity for both training sessions. Mean ($\pm SD$) blood lactate concentrations ($n = 187$) during the training sessions were 5.2 ± 2 and 7.2 ± 3 mmol·L⁻¹ for the HIIT and RST, respectively, with no significant differences between groups ($p = 0.053$).

DISCUSSION

To the best of our knowledge, this is the first study comparing the effects (i.e., physiological and performance adaptations) of 2 different high-intensity training protocols in competitive tennis players. The results show that both groups, HIIT and RST, showed similar improvements in aerobic fitness (e.g., $\dot{V}O_{2peak}$). However, the differences between groups were found with the HIIT-induced greater improvements in tennis-specific endurance, and the RST led to an improvement in RSA compared with HIIT. Although there is no single type of training recommended to improve all the factors responsible for performance in tennis, based on the present results, RST seems to represent a time-efficient stimulus for a simultaneous improvement of general and tennis-specific aerobic fitness as well for RSA.

Both training protocols, RST and HIIT, induced similar changes in aerobic fitness (e.g., increases in $\dot{V}O_{2peak}$ of 4.9 and 6.0%, respectively). Because of the lack of studies regarding training interventions in tennis, we are not able to compare our results with those of tennis players, for neither the interval training nor repeated sprint-based protocols. However, our results are consistent with those of previous studies using sport-specific games and running interval training protocols in soccer, with approximately 7% improvements in $\dot{V}O_{2max}$ after 4–8 weeks of training (16,21,29,42). After 6 weeks of training, we also found increases in the vLA4 for both groups, RST and HIIT, although the values were not significantly different from those of pretesting. There is a general consensus that $\dot{V}O_{2max}$ is limited mostly by the ability of the cardiovascular system to transport O₂ to active muscles, and it will be better adapted under high-intensity demands, lactate threshold is limited by the peripheral ability to use O₂ and, in particular, by mitochondrial enzyme activity (2,19,35). Although central factors (e.g., $\dot{V}O_{2peak}$) were restored rapidly in a relatively shorter time (4 weeks), peripheral factors (e.g., muscle oxidative enzymes) probably required a longer time to improve (a further 8 weeks) (3,24,29,32,37).

Comparing both training protocols, the players involved in the RST elicited similar improvements in aerobic fitness (i.e., $\dot{V}O_{2peak}$ and vLA4) than in the HIIT. In this regard, recent studies using sprint training have reported significant increases

in both anaerobic and aerobic power (22), which can be related to increases in glycolytic (38) and oxidative enzyme activities (11–13), muscle buffering capacity (4,18,19), and/or ionic regulation (12,45). The improvement in aerobic fitness after the RST protocol is consistent with the findings of previous studies using sprint-based training protocols (18,21,28,40,45). On using similar repeated-sprint training volumes as in Ferrari-Bravo et al. (21) (i.e., 220–240 m), we found significant improvements in selected parameters of aerobic fitness, such as $\dot{V}O_{2peak}$. This could be related to the impact on a quick oxidative recovery between the sprints and the decrease in the anaerobic demand during repeated sprints, with a subsequent inhibition of muscle glycolysis (e.g., blood lactate values obtained during RST training sessions were not different from the first to the last set of exercise [$p = 0.33$]) and contractile mechanisms during later sprints (i.e., and a performance decrease) (3,24,48).

The present results support the conclusions of Gibala et al. (24) that RSA-based training might be a time-efficient training strategy in enhancing aerobic adaptations, given the lower training volume required by the RST (~2.5 minutes per session, without active recovery) compared with that of the HIIT (13.5 minutes, without active recovery) in tennis players. During the past few years, tennis players have been observed to devote a great amount of time to improving their tennis skills throughout technical and tactical training, with an average of 15–20 hours of technical training per week even at a young age (16). Thus, from a practical point of view, desired adaptations (e.g., $\dot{V}O_{2peak}$ increases) can be obtained with a substantial reduction in exercise training time, allowing the players to spend more time on-court and optimizing technical and tactical skills (9,14,23,29). However, because the starting $\dot{V}O_{2peak}$ of the players in this study was close to the baseline limits reported in the literature (20,34), it is possible that a similar improvement in aerobic power was related to the low pretraining fitness level of the players involved in the study. Before our findings can be generalized to high performance tennis players, future studies should confirm our results with athletes characterized by a higher $\dot{V}O_{2peak}$.

This study included 2 different tennis-specific tests. The first one, more focused on the tennis-specific endurance (i.e., Hit and Turn Tennis Test), and the second one focused on the RSA (i.e., repeated-sprint test). Regarding the Hit and Turn test, the results showed significant increases in the maximum level achieved for both groups (i.e., an increase of 4 and 2 levels for HIIT and RST, respectively). Although the $\dot{V}O_{2peak}$ values for both groups were similar, the players involved in the HIIT showed greater improvements in the tennis-specific endurance test (Table 3). A possible explanation of this result could be that the Hit and Turn Test is designed to measure mainly the aerobic power of the subjects (20), in contrast with the RST, which can induce improvements in both aerobic and anaerobic metabolism (8,17,30). However, the main reason for this improvement seems to be that the HIIT followed a training program based on the Hit and Turn test (i.e., see Methods), which may have positively

influenced the performance from a neuromuscular point of view (e.g., footwork, stroke activity), leading to a more economic technical solution (8,9).

Regarding the RSA test, only the RST showed a significant reduction in the mean sprinting time (i.e., from 5.3 to 5.1 seconds). In addition, and for the HIIT, the improvement in the RSA performance of the RST may also be explained by the training protocol, because it is based on the RSA test. In this regard, the involvement of the same muscles during acceleration and deceleration movements (e.g., biceps femoris, rectus femoris, hip adductors, iliopsoas) could lead players to positive changes in specific coordination and agility during the RSA test (8,9). Moreover, and consistent with the findings of other authors, the repetition of short bouts of exercise stress not only many of the physiological or biochemical systems used in aerobic efforts (7,11–13,23,45), but it also induces alterations in glycolytic enzymes, muscle buffering, and ionic regulation resulting in improved anaerobic performance (4,18,26,38,40,43,46). This conclusion is supported by the observed decrease in the RSA mean time (i.e., reduction of 3.8%) concurrent with unchanged RSA decrement. This suggests an increase in the overall anaerobic performance but not in the ability to recover between sprints (21).

No effect of training protocols and no prechange to postchange were found in CMJ performance or 20-m sprint time, for neither HIIT nor RST (Table 3). Our results are in agreement with those of Ferrari-Bravo et al. (21) who found a lack of improvement in sprinting and jumping using 2 different training protocols (i.e., sprint vs. interval training) in soccer players. The lack of significant improvements in jumping height or sprint time in both training groups suggest that the observed improvements in RSA in this study were likely related to changes in specific coordination and agility (51), rather than because of enhancements in explosive force or sprinting mechanisms (46). Moreover, several studies have shown that qualities as speed and agility are independent locomotor skills and relatively unrelated to one another (39,50,51). This would lead to the assumption that straight speed and agility training methods produce limited transfer to the other directional running modes and suggest the use of a different combination of training methods (e.g., additional power and strength) to improve muscular power and hence short linear sprint ability. Moreover, this supports the idea that adaptations (e.g., mechanical, physiological) and associated changes in performance after a training intervention are to a certain extent training specific, with the energy system, the muscle group, the contraction force, or the movement patterns engaged, each playing a role in determining the final adaptations (8,9,36,44).

In conclusion, this study shows that both training programs used (i.e., RSA-based training and HIIT) might be appropriate to optimize the development of cardiorespiratory fitness in competitive tennis players, although the HIIT shows a greater improvement in the tennis-specific endurance (i.e., Hit and

Turn test), and only the RST induces improvements in RSA. The above discussion highlights the fact that there is not one single type of training that can be recommended to improve performance in tennis players. However, in terms of practicability, it seems that RSA-based training might be a time-efficient training strategy in enhancing aerobic adaptations, given the better adaptations (i.e., tennis-specific endurance and RSA) and the lower training volume required compared with the HIIT. However, the evaluation of combining effects and the participation of higher level tennis players (e.g., athletes characterized by a higher $\dot{V}O_{2peak}$) or the introduction of technical/tactical efficiency during the testing procedures warrants future studies.

PRACTICAL APPLICATIONS

The inclusion, 3 times per week, of either HIIT or RST programs to normal tennis training sessions represents an effective means of increasing performance-related physical fitness traits in high-level tennis players. The HIIT (i.e., $3 \times [3 \times 90 \text{ seconds}, 90\text{--}95\% \text{ HR}_{max}]$, with 3-minute rest) induced greater improvements in tennis-specific endurance and RST (i.e., $3 \times [10 \times 5\text{-second}]$ shuttle sprints, with 20-second rest between repetitions, 3 minutes between sets) led to a significant improvement in RSA. Despite the proven efficiency of the RST in improving specific qualities for tennis players (i.e., RSA), from a practical point of view, a combination of different training strategies seems to be more effective because several physiological systems are involved during tennis. The complementary use of HIIT exercises (e.g., work and rest intervals ranging from 15 second to 4 minutes; 90–100% velocity at the level of $\dot{V}O_{2max}$; HR values $>90\% \text{ HR}_{max}$; work to rest ratios of 1:1–4:1) should also be considered as training alternatives (10,33,35,37,52). The improvements in different qualities (e.g., RSA or $\dot{V}O_{2peak}$) in response to the exercise programs tested in this study illustrate the concept of training specificity and suggest that both training contents could be part of the training program in tennis players (44). These training-specific adaptations offer coaches and practitioners the possibility to individualize training content specific to the athletic qualities in tennis.

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